Integrated Approach Documentation

Pipeline

Phase 1: Set parameters and acquire data

1a: Set parameters

`M_0`: number of individuals to use for the first round of training (Phase 2)

`M`: number of infections to add tested samples to in a single round of iterative fitting (Phase 4)

`N`: number of tested samples to add to each of the `M` infections (Phase 4)

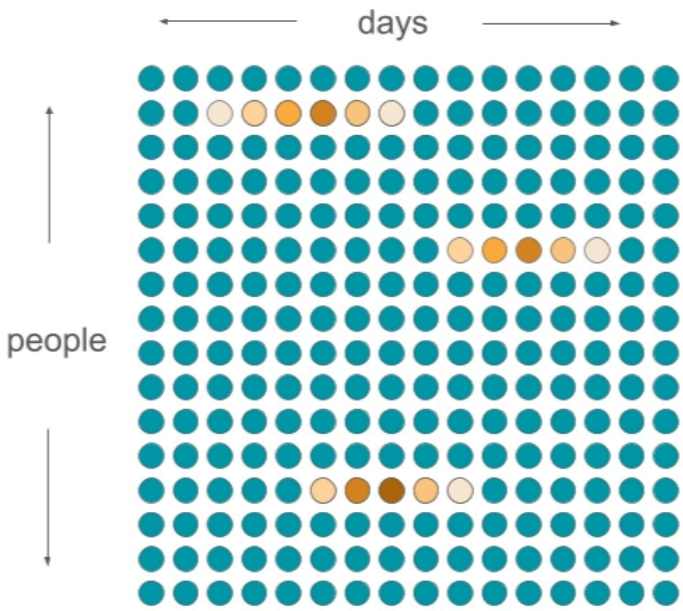
1b: Acquire data

read in `matrix`

dimensions: people-by-days

entries:

- equal to 0 when the sample corresponding to that person and day tests negative
- equal to some real number >0 (the viral RNA concentration) when the sample tests positive



`matrix` is our ground truth. In the following phases, we can't peek at an entry of the matrix without using a test.

Phase 2: Gather initial data

2a: Completely observe the first `M_0` rows of `matrix`

Test every entry in the first `M_0` rows of `matrix`

Output a dataset called `init_trajectories` whose rows are each a sample from a given person at a given time and whose columns are

- `index_values`: the index of the infection this sample is a part of, relative to the list of other infections discovered in the first `M_0` rows
- `time_values`: the time of this sample relative to the start of the infection
- `traj_values`: the viral RNA concentration of this sample
- `person_values`: the row in `matrix` corresponding to the individual this sample was taken from
- `date_values`: the column in `matrix` corresponding to the date on which this sample was taken

2b: Partially observe the remaining rows of the matrix

Set `l_0` (approximate lower bound on infection length)

Apply our pooling approach (described in these slides) to the remaining matrix rows

Output a dataset called `partial_trajectories` which has the same columns as `init_trajectories` but contains rows for only some of the sampling times of each trajectory

2c: Combine the complete and partial observations into one dataset

Stack `init_trajectories` and `partial_trajectories`

Output the combined dataset `all_trajectories`

Phase 3: Initial BHM fit

Fit the BHM (specified in the slides) to `all_trajectories`

Phase 4: Iterative testing and fitting

while the stopping criterion (currently an iteration limit) is not met:

- for the next `M` partially sampled trajectories:
  - test the `N` untested samples who have the highest Shannon entropy
  - for each test, add a row to `all_trajectories`
- fit the BHM to `all_trajectories`

Next Steps

- Plot RMSE over the iterations in Phase 4
- In Phase 3, only fit to `init_trajectories` , then iteratively add partially observed trajectories in Phase 4
  - this would require using our BHM to compute entropies for trajectories that the BHM has not yet seen

Code

https://github.com/Violet-Ross/efficient-sampling/tree/main/scripts/integrated_approach